

- [CPSC 375: High-Performance Computing](#)

[CPSC 375: High-Performance Computing](#)

Spring 2026

- [Syllabus](#)
- [Schedule](#)
- [Upload](#)
- [The Cluster Project](#)

Homework 11

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

- Consider the example program in Slides 13-14 of Lecture 11.
 - Fill each entry in the following table with “Yes” or “No” for the program. In the first column, the notation $v.t$ denotes an instance of variable v residing on the local stack for thread t , where t is either m (main thread), $p0$ (peer thread 0), or $p1$ (peer thread 1).

Variable instance	Referenced by main thread?	Referenced by peer thread 0?	Referenced by peer thread 1?
ptr			
cnt			
i.m			
msgs.m			
myid.p0			
myid.p1			

- Given the analysis in Part A, which of the variables ptr, cnt, i, msgs, and myid are shared?

- Complete the table for the following instruction ordering of badcnt.c (Slide 19 of Lecture 11):

Step	Thread	Instr	%eax ₁	%eax ₂	cnt
1	1	H_1	—	—	0
2	1	L_1	—	—	—
3	2	H_2	—	—	—
4	2	L_2	—	—	—
5	2	U_2	—	—	—
6	2	S_2	—	—	—
7	1	U_1	—	—	—
8	1	S_1	—	—	—
9	1	T_1	—	—	—
10	2	T_2	—	—	—

Does this ordering result in a correct value for cnt?

- **Welcome: Sean**

- [LogOut](#)

